Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Agentic AI engineer with production multi-agent systems: **Leasel** (8-agent LangGraph with eval harness, guardrails, HITL, and enterprise RAG - 93.6% precision) and **Magizhs** (LangGraph mission planner with planning/execution/verification loops over regulatory corpus). Understands how agents fail - built regression gating and LLM-as-judge eval harnesses that block failures before production. 8+ years shipping production systems at enterprise scale (100K+ MAU, 1.6M financial records). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Multi-Agent Orchestration | LangGraph / LangChain | RAG Pipelines & Memory Systems | Planning, Execution & Verification Loops | Guardrails & Reliability | Tool Calling & API Orchestration | Inference Optimization & Model Routing | Eval Harnesses & Agent Testing | Enterprise AI Agents | Production AI Systems

WORK EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI Agent Engineer (Intern)

- **Designed and deployed an end-to-end agentic system** for enterprise drone mission planning - LangGraph stateful graph with planning, execution, and verification loops (parse → plan → retrieve → validate → conditional routing) delivering risk-aware compliance strategy across marketing, real estate, and commercial survey verticals.
- RAG layer over FAA Part 107 regulatory corpus with cited-source outputs; validation guardrails enforcing altitude limits and no-fly-zone checks; self-directed from requirements through production with no spec doc.

George Mason University - College of Engineering
Aug 2025 - May 2026

Graduate Teaching Assistant

- Delivered technical workshops to 600+ students - Python programming, AI fundamentals, database design; translated complex AI architectures into hands-on exercises.

George Mason University - Research
Aug 2024 - Aug 2025

Research Associate - AI & Data Analytics

- Built multi-source prediction models (Python, R) across 190+ countries - R^2=0.977; Isolation Forest anomaly detection across 3 research pipelines.

Entrata - PropTech / Real Estate SaaS (100K+ MAU)
Sep 2018 - Aug 2024

Senior Software Engineer

- **Shipped 20+ full-stack production modules** (PHP, Python, React, PostgreSQL, RabbitMQ, AWS) across 120+ Agile sprints for 100K+ monthly active users - 99.5% uptime; led microservices redesign (Docker) with 50% scalability improvement and 30% deployment time reduction.
- **Managed end-to-end financial services functionalities** - HAP allocation, AMI-based pricing, HUD compliance reporting, LIHTC recertification workflows; zero compliance violations across 100K+ unit portfolios.
- **Optimized 200+ SQL queries** (30% performance gain); 100% PHI compliance on AWS S3 migration with Libsodium-encrypted pipelines; eliminated 80% of manual workload via Python/Bash automation.
- 300+ peer code reviews; mentored 5+ junior engineers; defined data standards and integration patterns across 20+ production modules.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built full-stack property management modules (PHP, Python, PostgreSQL); ETL pipelines on 1.6M records; AWS S3 cloud migration.

PRODUCTION AGENTIC SYSTEMS

Leasel: 8-Agent Enterprise AI Platform - Guardrails, Eval Harness, RAG 2026

8-agent LangGraph StateGraph with real conditional routing and planning/execution/verification loops: Triage → Retrieval → Configuration → Historical → Reasoning → Validation → HITL Escalation. **Guardrails enforced in code**: LLM reasons; deterministic logic makes every regulated call; confidence scoring triggers HITL when uncertain; 100% citation correctness; audit trail on every decision. **RAG pipeline**: section-aware chunking, TF-IDF retrieval, 91% Precision@1. **Eval harness**: 225-query benchmark, pass@k multi-trial sampling, LLM-as-judge, regression gating - validates in simulation before every release. Production precision: 93.6% across 1,240 enterprise documents.

Python · LangGraph · FastAPI · RAG · Tool Calling · HITL · Guardrails · Eval Harnesses · LLM-as-Judge · Regression Gating

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia — May 2026

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University — 2017

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, PHP, R, Bash

Agentic AI: LangChain, LangGraph, Multi-Agent Orchestration, Enterprise AI Agents, Tool Calling, API Orchestration, Planning Loops, Execution Loops, Verification Loops, RAG Pipelines, Memory Systems, Embeddings, Vector Databases, ChromaDB, Guardrails, HITL, Confidence Scoring, Inference Optimization, Model Routing, Eval Harnesses, LLM-as-Judge, Regression Gating, MCP, OpenAI API, Anthropic API, Groq, Ollama, FastAPI, spaCy, BERT

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, CI/CD (GitHub Actions), PostgreSQL, MySQL, Oracle, RabbitMQ, Salesforce, Git, Linux, Agile/Scrum

Data & ML: PyTorch, Scikit-learn, XGBoost, Isolation Forest, Pandas, NumPy, Tableau, Power BI